

Phishing Email Analysis Project

Project Overview

This project demonstrates phishing email investigation using email header analysis and threat intelligence tools. The goal was to identify phishing indicators such as spoofed sender domains, SPF/DKIM/DMARC failures, suspicious IP addresses, and malicious email behavior.

Tools Used

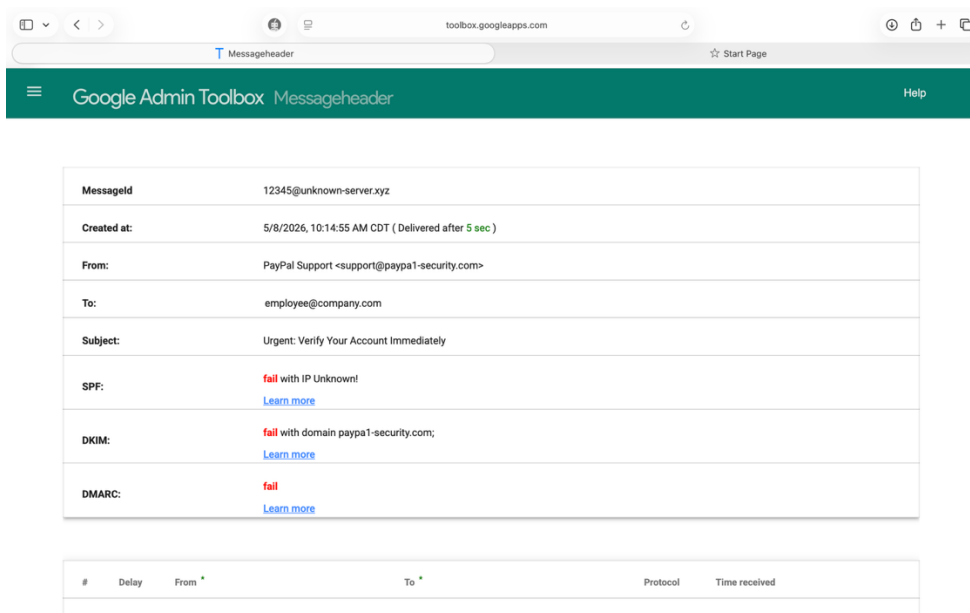
- Google Admin Toolbox Messageheader
- VirusTotal
- AbuseIPDB
- MXToolbox

1. Email Header Analysis

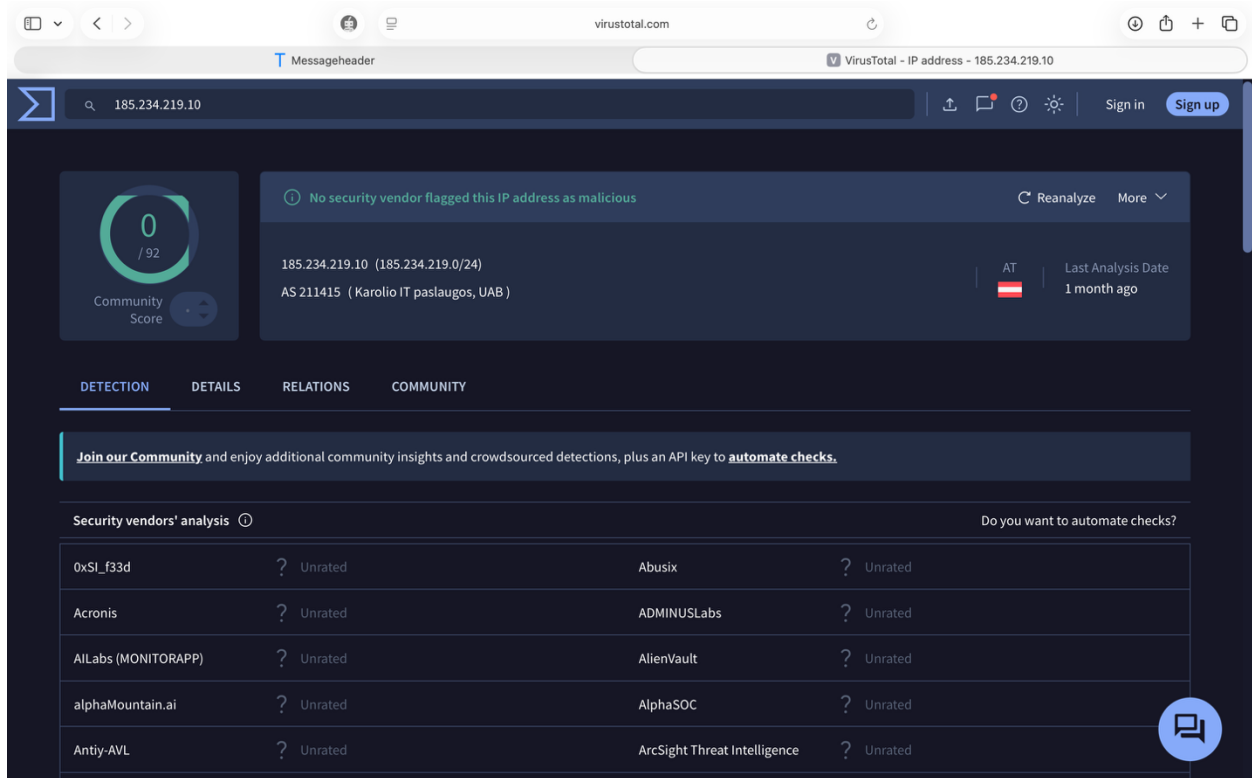
This screenshot shows the phishing email investigation using Google Admin Toolbox Message header.

Findings:

- Fake sender domain detected: paypal-security.com
- SPF = Fail
- DKIM = Fail
- DMARC = Fail
- Suspicious sender infrastructure detected



2. Virus Total IP Analysis



The screenshot shows the VirusTotal website interface for an IP address analysis. The browser address bar shows 'virustotal.com' and the page title is 'VirusTotal - IP address - 185.234.219.10'. The main content area displays the IP address '185.234.219.10' and its details: 'AS 211415 (Karolio IT paslaugos, UAB)'. A 'Community Score' of 0/92 is shown. A message states 'No security vendor flagged this IP address as malicious'. Below this, there are tabs for 'DETECTION', 'DETAILS', 'RELATIONS', and 'COMMUNITY'. A section titled 'Security vendors' analysis' shows a table of vendors and their ratings. The table has two columns of vendors, each with a rating of 'Unrated'. A button 'Reanalyze' and a 'More' dropdown are visible. A 'Sign in' button and a 'Sign up' button are in the top right corner. A blue chat bubble icon is in the bottom right corner.

185.234.219.10

0 / 92
Community Score

No security vendor flagged this IP address as malicious

185.234.219.10 (185.234.219.0/24)
AS 211415 (Karolio IT paslaugos, UAB)

AT Last Analysis Date
1 month ago

Reanalyze More

DETECTION DETAILS RELATIONS COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis Do you want to automate checks?

0xSI_f33d	? Unrated	Abusix	? Unrated
Acronis	? Unrated	ADMINUSLabs	? Unrated
AILabs (MONITORAPP)	? Unrated	AlienVault	? Unrated
alphaMountain.ai	? Unrated	AlphaSOC	? Unrated
Antiy-AVL	? Unrated	ArcSight Threat Intelligence	? Unrated

IP Investigated: 185.234.219.10

Findings:

- 0/92 vendors flagged as malicious
- IP currently unrated
- Community score = 0
- Likely low-reputation hosting infrastructure used in phishing activity

Upload your screenshot here:

- virustotal-analysis.png

Indicators of Compromise (IOCs)

Type	Value
Domain	paypal-security.com
IP Address	185.234.219.10
Sender Email	support@paypal-security.com
Subject	Urgent: Verify Your Account Immediately

Incident Summary

A phishing email impersonating PayPal support was analyzed using header inspection and threat intelligence validation.

Key Findings

- Unauthorized sender detected through SPF failure
- Email authenticity failed through DKIM failure
- Domain spoofing confirmed through DMARC failure
- Suspicious sender domain impersonating PayPal
- Sender IP investigated using VirusTotal

Response Recommendations

- Block malicious domain and sender
- Monitor suspicious IP activity
- Educate users on phishing indicators
- Strengthen email security policies
- Enable advanced phishing detection rules

Conclusion

This project demonstrates practical SOC Analyst skills in phishing detection, email header analysis, IOC extraction, and incident reporting. It supports internship applications for SOC Analyst, Cybersecurity Analyst, and Information Security roles.